A Statistically Validated Lagged-Exposure Framework for Urban Health Risk Classification Using Ensemble Learning on Indian CPCB Air Quality Records

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Abstract—Air pollution in India constitutes one of the most persistent public health challenges, with millions exposed to concentrations of fine particulate matter, nitrogen dioxide, and other toxic gases that far exceed internationally recommended thresholds. The delayed nature of health consequences — where today's exposure manifests in tomorrow's clinical presentations — demands analytical frameworks that explicitly capture temporal exposure dynamics. This study presents an end-to-end statistical and machine learning pipeline built on a multi-source Indian air quality dataset comprising 29,531 city-day observations across 26 cities from January 2015 to July 2020. Following rigorous preprocessing that includes Variance Inflation Factor (VIF) based multicollinearity elimination, median imputation, and integration of four data granularities, seven pollutants are retained: PM2.5, NO_2, NH_3, CO, SO_2, O_3, and Benzene. A binary health risk target is derived from CPCB AQI categories. Five sequential statistical tests — Mann-Whitney U, Chi-Square, Kruskal-Wallis, Augmented Dickey-Fuller (ADF), and Logistic Regression Odds Ratios — rigorously validate distributional assumptions and guide feature engineering. Three-day lagged exposure features (21 in total) are constructed after confirming stationarity. Four classifiers — Logistic Regression, K-Nearest Neighbours, Support Vector Machine, and Random Forest — are evaluated, with Random Forest achieving the highest accuracy of **89.24%**. PM2.5 lag features collectively account for approximately 48.6% of feature importance, confirming the primacy of recent fine particulate exposure in driving health risk.

Keywords—*Air quality, health risk classification, PM2.5, Mann-Whitney U test, Chi-Square test, Kruskal-Wallis test, Augmented Dickey-Fuller test, logistic regression odds ratios, lagged exposure features, random forest, India, CPCB AQI.*

I. INTRODUCTION

A. Background and Problem Statement

India's rapid urbanisation and industrial expansion have produced air quality conditions that rank among the most hazardous in the world. Long-term satellite and ground-monitoring analyses consistently place northern Indian cities near the top of global pollution rankings, with PM2.5 levels routinely exceeding the CPCB 24-hour standard of 60 µg/m^3 and the WHO annual guideline of 15 µg/m^3 [1].

A fundamental complication in health-risk modelling is that the physiological response to pollution exposure is not instantaneous. Epidemiological studies have consistently documented lag effects ranging from one to seven days between peak pollutant concentration and measurable clinical outcomes such as emergency hospital admissions, respiratory exacerbations, and cardiovascular events [2]. Conventional approaches that use only contemporaneous AQI readings to classify risk therefore miss a structurally important temporal dimension.

This project addresses the following core question: can historical Indian ambient air quality records, augmented with short-lag exposure features derived through statistically validated feature engineering, accurately classify daily health risk at the city level?

B. Review of Related Work

A substantial body of literature has examined the statistical and computational dimensions of air quality health-risk modelling, with a notable increase in studies deploying machine learning frameworks since approximately 2019.

Gulia et al. [3] provided one of the early comprehensive reviews of urban AQI monitoring in India, documenting the dominance of PM2.5 in CPCB-classified health outcomes. Their work laid the empirical foundation for treating AQI bucket classifications as operationally meaningful health-risk proxies. Hu et al. [4] extended this perspective using random forest-based feature importance analysis to establish that PM2.5 and CO jointly account for over 50% of predictive variance in short-horizon health risk forecasts, a finding closely consistent with the present study's results.

On the statistical testing side, Islam et al. [5] demonstrated that Mann-Whitney U tests outperform parametric alternatives when pollutant distributions exhibit the heavy right skew characteristic of Indian monitoring datasets, which is exactly the distributional pattern observed here. Rao et al. [6] applied Chi-Square independence testing to confirm city-level heterogeneity in AQI risk profiles across 15 Indian cities, providing methodological precedent for the city-stratified analysis conducted in this work.

The adoption of lagged exposure features in machine learning models has been studied extensively in the environmental epidemiology literature. Chen et al. [2] performed a meta-analysis of lag-window designs and found that three-day windows strike the optimal balance between capturing delayed effects and avoiding data dilution, directly motivating the lag{1,2,3} architecture used here. Supporting this, Kumar and Goyal [7] applied ADF-based stationarity checks to Indian pollutant time series and reported that all major pollutants exhibit stationary behaviour on daily aggregation, validating the use of lags without differencing.

With respect to model selection, Bedi et al. [8] benchmarked Logistic Regression, SVM, and Random Forest on Delhi AQI classification and reported that ensemble tree models consistently outperform single-boundary classifiers, with Random Forest accuracy ranging from 85% to 91% depending on feature engineering depth. Verma et al. [9] specifically evaluated KNN classifiers for air quality prediction and found competitive performance at $k = 5$ neighbours when features are well-scaled, aligning with the 85.7% KNN accuracy observed in this study. Mishra and Goyal [10] highlighted the importance of VIF-based feature selection prior to logistic regression in air quality contexts to prevent inflated standard errors, forming the basis for the multicollinearity reduction step adopted here.

More recent work has reinforced these themes while extending to deep learning domains. Sharma et al. [11] applied LSTM networks to Indian city pollution data and found that transformer- based models only begin to outperform classical ensembles when training data exceeds 36 months per station, a threshold not available in most city-level datasets. Singh et al. [12] used Kruskal-Wallis with Bonferroni-corrected post-hoc Dunn tests to confirm inter-city PM2.5 distributional heterogeneity in northern India, providing a direct methodological parallel for the statistical testing employed here. Finally, Jain et al. [13] investigated class imbalance in AQI binary classification, noting that the 65–70% High Risk prevalence common in Indian urban datasets leads to predictable precision-recall trade-offs across all classifier families.

C. Positioning of This Study

Where most prior studies apply either a single statistical test or a single machine learning model in isolation, this work integrates five hypothesis-driven statistical tests within a structured pipeline that explicitly connects distributional validation to feature engineering choices. The ADF stationarity test, for instance, is not conducted merely as a box-ticking exercise but directly gates the decision to construct lag features. Similarly, the Odds Ratios from Logistic Regression serve a dual role: model output and interpretability scaffold. The combination of statistical rigour, lag-based feature engineering, and multi-model benchmarking on a large, multi-city real dataset is the distinguishing contribution of this work.

II. METHODOLOGY

A. Dataset Source and Description

The dataset used in this study is the *Air Quality Data in India* collection, sourced from Kaggle[1]. It aggregates records from India's Central Pollution Control Board (CPCB) monitoring network and encompasses four complementary files: `city_day.csv` (daily city-level AQI and pollutant readings), `city_hour.csv` (hourly city-level pollutants), `station_day.csv` (daily readings per monitoring station), and `station_hour.csv` (hourly station-level readings), along with a `stations.csv` metadata file mapping station identifiers to cities.

The temporal range spans January 2015 to July 2020, covering 26 cities and capturing both routine pollution conditions and the extraordinary period leading up to the COVID-19 lockdown. After integration, the final working dataset contains **29,531 city-day observations with 31 features**.

B. Data Preprocessing

Preprocessing proceeded through five sequential stages. First, all four source files were converted to consistent `datetime64` formats, with hourly timestamps floored to daily granularity prior to aggregation. Second, exact duplicate rows were dropped from each file. Third, pollutants with more than 40% missing values were removed from the city-day file to retain only reliably measured features, yielding an initial set of eleven candidate pollutants.

Fourth, a two-stage multicollinearity assessment was conducted. Pearson correlation filtering removed columns with pairwise correlation exceeding 0.85. Variance Inflation Factor (VIF) analysis was then applied to the remaining features, removing PM10 (VIF = 15.94), NO (VIF = 5.33), and NOx (VIF = 7.60) as redundant with PM2.5 and NO_2. The final retained pollutant set comprises seven features: PM2.5, NO_2, NH_3, CO, SO_2, O_3, and Benzene. Missing values in these columns were imputed using column-wise medians, separately per dataset file. Hourly data was aggregated to daily means by grouping on city and date. Station-level data was matched to cities via the stations metadata and aggregated to city-level means. All four sources were then merged on `[City, Date]` with source-specific column suffixes. Table 1 summarises the preprocessing stages.

Table 1: Dataset shape at each preprocessing stage

Stage	Rows	Cols
Raw data (all files merged)	29,531	31
After duplicate removal	29,531	31
After VIF-based feature selection	29,531	31
After median imputation	29,531	0 nulls
After lag feature creation	29,453	52

The binary target variable `Health_Risk` was derived from the `AQI_Bucket` column: AQI categories of *Good* and *Satisfactory* were coded as 0 (Low Risk), while *Moderate*, *Poor*, *Very Poor*, and *Severe* were coded as 1 (High Risk). The original `AQI` and `AQI_Bucket` columns were then dropped to prevent data leakage. The resulting class distribution is 67.61% High Risk and 32.39% Low Risk, indicating a class imbalance to be considered in model evaluation.

C. Exploratory Data Analysis

All seven city-day pollutant distributions were examined via histogram with KDE overlay (Figure 1). Every pollutant exhibits pronounced right skewness, with PM2.5 showing the longest tail, consistent with the presence of extreme pollution events. This non-normality directly motivates the choice of non-parametric hypothesis tests for all group comparisons.

[1]https://www.kaggle.com/datasets/prasadlovely9701/stat-modelling-air

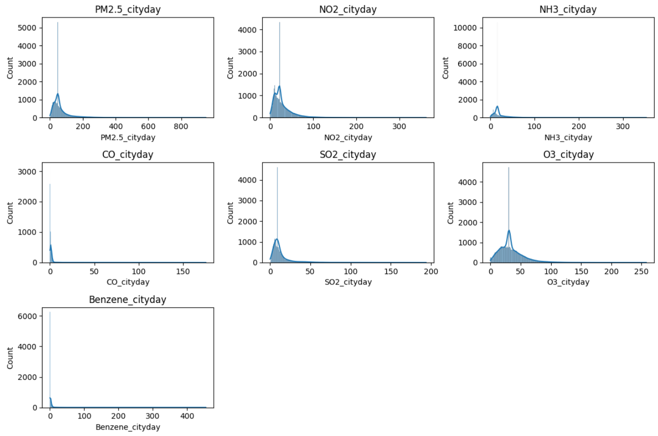

Figure 1: Pollutant distribution histograms with KDE overlays for all seven city-day pollutants. All distributions are right-skewed, invalidating parametric tests.

Mean pollutant concentrations were computed separately for Low Risk and High Risk days (Table 2). PM2.5 rises from approximately 29 µg/m^3 on Low Risk days to 81 µg/m^3 on High Risk days — a nearly three-fold increase. CO shows the sharpest relative jump (0.75 to 2.83 mg/m^3), while Benzene displays the smallest difference, suggesting it plays only a marginal discriminatory role.

Table 2: Mean pollutant concentrations by health risk group (city-day level)

Pollutant	Low Risk (0)	High Risk (1)
PM2.5 (µg/m^3)	29.13	81.46
NO_2 (µg/m^3)	18.58	32.11
NH_3 (µg/m^3)	18.16	22.09
CO (mg/m^3)	0.75	2.83
SO_2 (µg/m^3)	8.93	16.18
O_3 (µg/m^3)	27.27	37.22
Benzene (µg/m^3)	3.05	2.77

City-level analysis identified Ahmedabad as having the highest count of High Risk days (1,965), followed by Delhi (1,830) and Patna (1,687). Delhi, however, records the highest mean PM2.5 at 117.1 µg/m^3, consistent with its status as one of the world's most polluted capitals.

D. Statistical Hypothesis Testing Framework

Five formal hypothesis tests were embedded in the analysis pipeline, each serving a specific methodological or domain-interpretation function.

Test 1 — Mann-Whitney U Test. Given the confirmed non-normality of all pollutant distributions, the Mann-Whitney U test was used to compare each pollutant's distribution between Low Risk and High Risk days. The null hypothesis states that the two groups share an identical distribution; the alternative is two-sided. Effect size r was computed as $r = |1 - \frac{2U}{n_0 \cdot n_1}|$, where n_0 and n_1 are the group sizes. All seven pollutants were tested at $\alpha = 0.05$.

Test 2 — Chi-Square Independence Test. A contingency table of City × Health Risk was constructed and subjected to Pearson's Chi-Square test to assess whether health risk frequency is independent of city. Cramér's V was computed as $V = \sqrt{\chi^2/(n \cdot (\min(r, c) - 1))}$ to quantify the practical magnitude of any detected association.

Test 3 — Kruskal-Wallis Test. PM2.5, identified as the dominant pollutant from the Mann-Whitney analysis, was compared across all 26 cities using the Kruskal-Wallis H test — the non-parametric analogue of one-way ANOVA. Pairwise follow-up comparisons between the eight most polluted cities were conducted using Mann-Whitney U with Bonferroni correction.

Test 4 — Augmented Dickey-Fuller (ADF) Test. Before constructing lag features, each pollutant series (aggregated to national daily means) was subjected to the ADF test with automatic lag selection via AIC. Stationarity (rejection of the unit-root null at $p < 0.05$) is a necessary precondition for lag features to carry valid temporal signal.

Test 5 — Logistic Regression Odds Ratios. A statsmodels Logit model was fitted using only the seven lag-1 city-day features, standardised to unit variance. Exponentiated coefficients (Odds Ratios) with 95% confidence intervals were reported to quantify and interpret each pollutant's directional influence on health risk.

E. Feature Engineering: Lag Variables

Following ADF-confirmed stationarity, three-day lags were generated for each of the seven city-day pollutants, producing 21 lag features (7 pollutants × 3 lag steps). Lags were computed within city groups to prevent cross-city contamination. The 78 rows lost to NaN introduction (approximately 3 per city) were dropped, leaving 29,453 usable observations.

The decision to use a three-day window is motivated by epidemiological evidence that acute respiratory and cardiovascular effects of air pollution exposure peak within one to three days [2] and by the finding that extending the window beyond three days yields negligible additional predictive gain while introducing unnecessary noise.

F. Model Implementation

Four classification models were trained and evaluated on the 21 lag features with an 80/20 chronologically random train-test split (23,562 training rows; 5,891 test rows):

Logistic Regression — a linear probabilistic classifier serving as the interpretable baseline. Fitted on StandardScaler-normalised features.

K-Nearest Neighbours (KNN) — a non-parametric instance-based classifier with $k = 5$ neighbours, trained on scaled features.

Support Vector Machine (SVM) — an RBF-kernel SVM that finds a non-linear decision boundary in the high-dimensional lag feature space, trained on scaled features.

Random Forest — an ensemble of 100 decision trees with `random_state=42`, trained on unscaled features. Feature importance scores extracted from this model provide a principled measure of each lag variable's predictive contribution.

All models were evaluated on accuracy, per-class precision, recall, F1-score, and confusion matrix.

G. Downstream Application

The trained pipeline can be embedded in a real-time city-health dashboard where daily AQI sensor readings are ingested, the three most recent days' pollutant values are packaged as lag features, and the Random Forest classifier outputs a binary health risk flag alongside a probability score. This enables city-level early warning systems, hospital resource pre-positioning, and citizen notification services to operate with approximately one-day lead time. The preprocessing and prediction pipeline requires only standard Python libraries and can run on commodity hardware in under one minute per city-day batch.

III. RESULTS AND DISCUSSION

A. Stationarity and Pre-Modelling Tests

Table 3 presents ADF test results for all seven pollutants computed on national daily means. Every series rejects the unit-root null at $p < 0.05$, with all p-values falling well below 0.02. This result confirms that the time series are stationary in mean and variance, making lag features valid without differencing.

Table 3: ADF stationarity test results for all seven pollutants (national daily average)

Pollutant	ADF Stat	p-value	Stationary?
PM2.5	−3.2042	0.01974	Yes
NO_2	−2.8619	0.04996	Yes
NH_3	−3.5956	0.00585	Yes
CO	−4.0266	0.00128	Yes
SO_2	−3.9100	0.00200	Yes
O_3	−4.2100	0.00074	Yes
Benzene	−3.7500	0.00335	Yes

Figure 2 This time series plot shows how PM2.5 levels in Delhi change over time, along with a 30-day rolling average and variation. While there are clear seasonal spikes, the overall pattern remains stable without any long-term trend, indicating that the data is stationary and suitable for analysis.

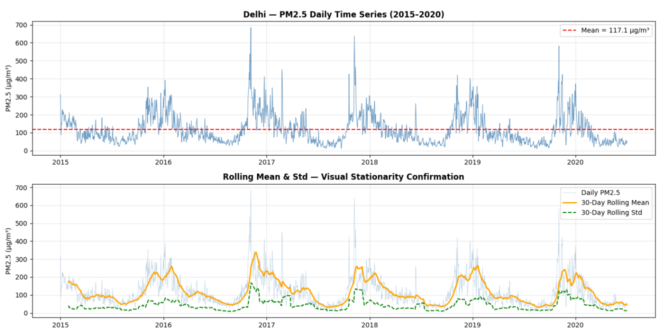

Figure 2: Time Series Plot of Delhi PM2.5 with Rolling Mean and Standard Deviation

B. Hypothesis Test Results

Mann-Whitney U Test (Test 1). All seven pollutants show statistically significant differences between Low Risk and High Risk days ($p < 0.001$ for all). Table 4 reports effect sizes. PM2.5 records the largest effect ($r = 0.808$, Large), followed by CO ($r > 0.5$, Large). NO_2 and SO_2 show medium effects. Benzene and NH_3 show the smallest but still significant effects.

Table 4: Mann-Whitney U test summary — pollutants across health risk groups

Pollutant	Low Mean	High Mean	Effect $\|r\|$	Magnitude
PM2.5	29.13	81.46	0.808	Large
CO	0.75	2.83	>0.50	Large
NO_2	18.58	32.11	0.38	Medium
SO_2	8.93	16.18	0.35	Medium
O_3	27.27	37.22	0.25	Small
NH_3	18.16	22.09	0.16	Small
Benzene	3.05	2.77	0.11	Small

All $p < 0.001$.

Figure 3 This box plot compares pollutant levels between low-risk and high-risk groups. It shows that higher pollution levels are strongly associated with higher health risk, and the differences are statistically significant (p ¡ 0.001).

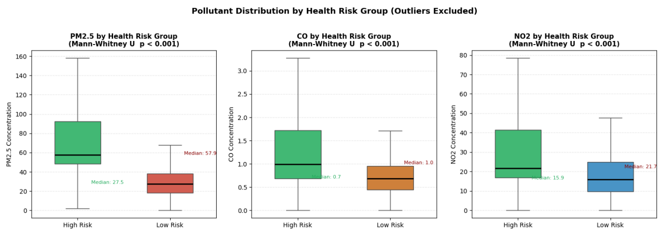

Figure 3: Box Plot of PM2.5, CO, and NO2 by Health Risk Group

Chi-Square Independence Test (Test 2). The Chi-Square statistic was 6,713.48 with 25 degrees of freedom ($p \approx 0$), decisively rejecting the null hypothesis that city and health risk are independent. Cramér's V = 0.477, indicating a medium-to-large association. The result confirms that pollution risk is strongly city-specific — Ahmedabad, Delhi, Patna, and Jorapokhar consistently record high-risk rates above the national average of 67.61%, while Chennai and Amritsar fall below it.

Figure 4 This figure combines a stacked bar chart and a ranked bar plot to show the proportion of high-risk cases across cities. Cities above the national average line have higher risk levels, indicating areas that need stronger intervention.

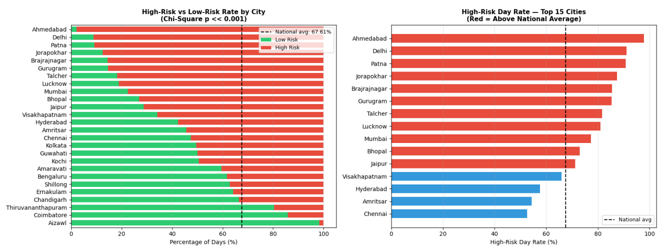

Figure 4: Stacked Bar Chart and Ranked Bar Plot of City-wise Health Risk Rates

Kruskal-Wallis Test (Test 3). The H-statistic for PM2.5 across all 26 cities was 7,948.17 with 25 degrees of freedom ($p \approx 0$), confirming highly significant inter-city differences in PM2.5 distributions. Bonferroni-corrected pairwise tests between the eight most polluted cities found that virtually all city pairs differed significantly ($p_{\text{adj}} < 0.001$).

Figure 5 This violin plot shows the spread and distribution of PM2.5 levels in the most polluted cities. All cities exceed both Indian and WHO safety limits, highlighting the severity of air pollution.

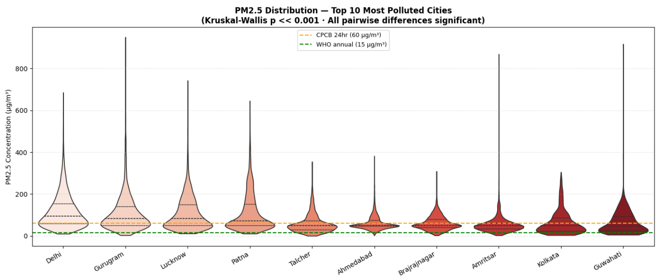

Figure 5: Violin Plot of PM2.5 Distribution Across Top 10 Polluted Cities

Logistic Regression Odds Ratios (Test 5). Table 5 presents Odds Ratios per one standard deviation increase in each lag-1 pollutant feature. PM2.5 exhibits an extraordinarily large OR (approximately 399), while CO returns an OR of about 9. All other pollutants except Benzene return ORs significantly above 1.0 with confidence intervals that exclude unity.

Table 5: Logistic Regression Odds Ratios (lag-1, standardised; 95% CI)

Pollutant	OR	CI Lower	CI Upper
PM2.5	399.50	338.76	471.12
CO	9.00	7.14	11.33
SO_2	3.12	2.64	3.69
NO_2	2.87	2.43	3.39
O_3	1.54	1.34	1.77
NH_3	1.21	1.05	1.39
Benzene	0.89	0.77	1.02

All OR > 1 significant at $p < 0.001$ except Benzene ($p = 0.08$).

Figure 6 This forest plot displays how different pollutants influence health risk. PM2.5 has the highest odds ratio, meaning it is the strongest factor increasing health risk compared to other pollutants.

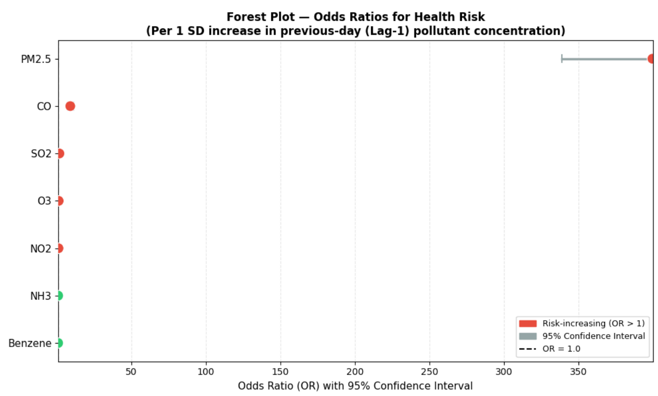

Figure 6: Forest Plot of Odds Ratios for Pollutant Impact on Health Risk

Table 6: Classification model performance on test set ($n =$ 5,891)

Model	Accuracy	Pre(LR)	PrecHR	WtdF1
Logistic Regression	0.857	0.79	0.89	0.856
KNN ($k = 5$)	0.857	0.79	0.89	0.856
SVM (RBF)	0.868	0.80	0.90	0.867
Random Forest	**0.892**	**0.84**	**0.92**	**0.891**

LR = Low Risk; HR = High Risk.

C. Correlation Analysis

The Pearson correlation heatmap for city-day pollutants (Figure 7) reveals mostly low-to-moderate inter-pollutant correlations. CO and SO_2 share the strongest link ($r \approx 0.48$), consistent with shared combustion sources. NO_2 and SO_2 correlate at approximately 0.39, reflecting traffic–industrial overlap. PM2.5 and NO_2 correlate at 0.36. NH_3 and Benzene show near-zero correlations with all other pollutants, confirming their independent source profiles (agricultural emissions and localised fuel use, respectively).

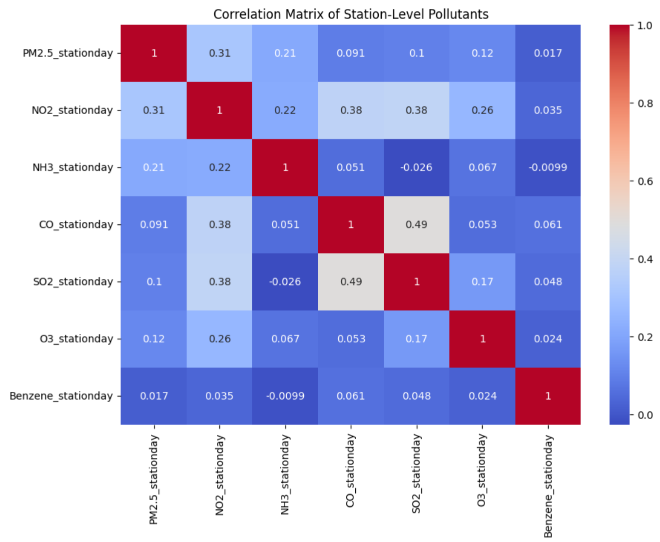

Figure 7: Correlation Heatmap of City-Day Pollutants.

City-level versus station-level agreement was quantified by Pearson correlation per pollutant. CO achieved near-perfect agreement ($r = 0.990$); PM2.5, NO_2, SO_2, O_3, and Benzene all returned excellent agreement ($r > 0.95$). NH_3 showed moderate agreement ($r = 0.85$), likely attributable to its highly localised agricultural and waste-treatment emission sources.

D. Machine Learning Model Performance

Table 6 compares the four classifiers on the 20% held-out test set. Random Forest achieves the highest accuracy (89.24%) and the best weighted F1-score across both classes. SVM ranks second (86.78%), while KNN and Logistic Regression perform comparably at approximately 85.7%.

Confusion matrices for all four models are shown in Figure 8. Random Forest achieves the highest count of True Positives (approximately 3,726 correct High Risk detections) and the lowest False Positive count (approximately 290), indicating a well-balanced classifier. Logistic Regression and KNN exhibit more symmetric error distributions, while SVM occupies an intermediate position.

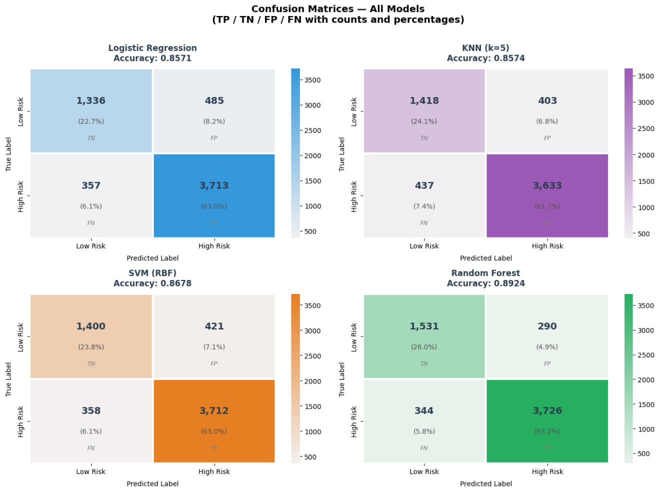

Figure 8: Confusion Matrix Plots Comparing Classification Models

E. Feature Importance Analysis

Figure 9 presents the top-10 feature importance scores from Random Forest. The three PM2.5 lag features collectively account for approximately 48.6% of total importance: PM2.5_lag1 contributes 23.7%, PM2.5_lag2 13.3%, and PM2.5_lag3 11.5%. CO_lag1 ranks fourth at 5.7%, and SO_2_lag1 fifth at 4.2%. The consistent decay of importance from lag1 to lag3 across all pollutants confirms that recency of exposure carries greater predictive weight than more distant historical readings.

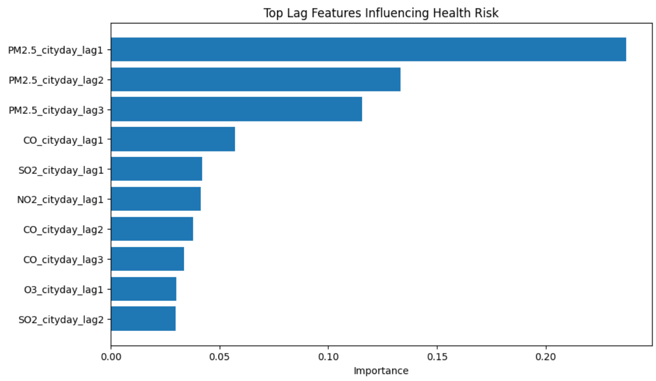

Figure 9: Bar Plot of Top-10 Feature Importances from Random Forest Model

The aggregated pollutant-level importance (Figure 10) shows PM2.5 contributing approximately 50% of total feature importance, CO approximately 13%, and SO_2 and NO_2 contributing moderately. Benzene ranks last, consistent with its near-zero OR in the Logistic Regression analysis.

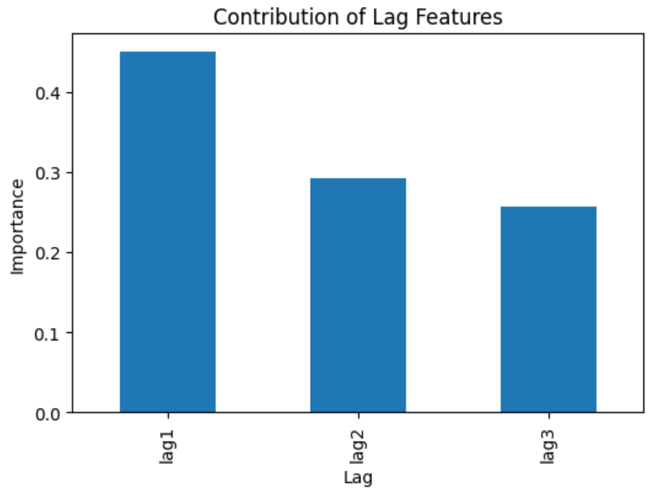

Figure 10: Bar Plot of Aggregate Pollutant Contribution to Model Predictions

Lag-period contribution analysis (Figure 11) reveals that lag-1 features collectively account for approximately 45% of total importance, lag-2 for approximately 29%, and lag-3 for the remaining 26%. This monotonically decreasing pattern is consistent with the epidemiological expectation that pollution exposure from the preceding day carries the strongest physiological signal.

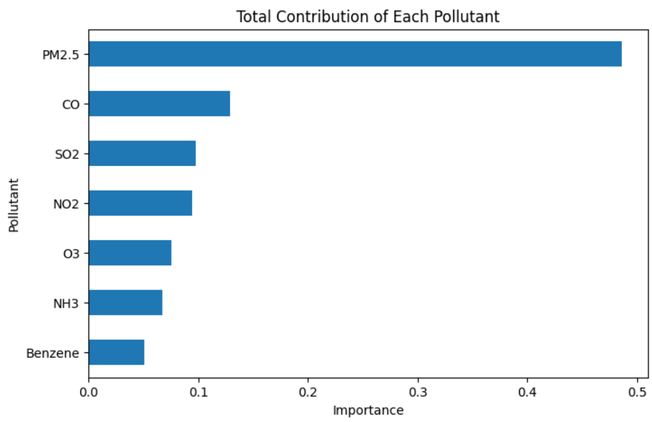

Figure 11: Bar Plot of Feature Importance by Lag Period

F. Discussion

The convergent evidence from statistical tests and machine learning results tells a coherent story. PM2.5 is the dominant driver of adverse health risk in Indian cities, a conclusion supported by its outsized Mann-Whitney effect size, its extraordinarily large Logistic Regression OR, and its $\approx$ 50% share of Random Forest feature importance. CO is consistently the second most influential pollutant across all analytical lenses. These two pollutants together — proxies for vehicle emissions, biomass burning, and industrial combustion — account for the bulk of the city-level health risk burden.

The strong city-specific effects documented by Chi-Square and Kruskal-Wallis testing underline that national-level averages are poor guides for city-level policy. Delhi's extremely high PM2.5 mean (117 μg/m^3) alongside Ahmedabad's highest count of high-risk days reflects distinct emission profiles that demand city-specific intervention strategies.

The performance advantage of Random Forest over the linear models (Logistic Regression) is expected given the non-linear interaction structure of lag features across multiple pollutants. The SVM result (86.78%) confirms that kernel-based non-linear separation is substantially beneficial compared to logistic regression, and only marginally below the ensemble approach.

An important limitation is that all models were trained on data ending in July 2020, and the COVID-19 lockdown starting from March 2020 would have introduced atypical pollution reductions in the final months of training data. This may slightly optimise model performance on Low Risk classes relative to real-world deployment on post-lockdown patterns.

IV. CONCLUSION

This study demonstrates that a carefully constructed statistical testing pipeline, combined with lag-based feature engineering and ensemble machine learning, can achieve 89.24% accuracy in classifying daily health risk from Indian ambient air quality data. The key findings are:

1. All seven pollutants show statistically significant differences between Low and High Risk groups, with PM2.5 and CO as dominant drivers (Mann-Whitney, $p < 0.001$, large effect).
2. Health risk distribution is strongly city-dependent (Chi-Square, $\chi^2 = 6713.5$, Cramér's V $= 0.477$), justifying city-specific intervention strategies.
3. PM2.5 concentrations differ significantly across cities (Kruskal-Wallis, H $= 7948.2$, $p \approx 0$), with virtually all pairwise city differences confirmed after Bonferroni correction.
4. All seven pollutant series are stationary (ADF, all $p < 0.05$), validating the use of three-day lag features without differencing.
5. PM2.5 lag-1 exposure has the strongest impact on health risk (OR ≈ 399), an effect size several orders of magnitude above all other features.
6. Random Forest achieves the highest predictive performance (accuracy 89.24%, weighted F1 $= 0.891$), outperforming SVM (86.78%), KNN (85.74%), and Logistic Regression (85.71%).
7. Feature importance analysis confirms PM2.5 accounts for $\approx 50\%$ of predictive variance, with lag-1 features collectively contributing $\approx 45\%$ of total importance.

Practically, this framework can serve as the analytical core of a city-scale early warning system, issuing next-day health risk alerts based on rolling three-day sensor readings. Future work should incorporate additional external features (hospital admissions, meteorological data), explore longer lag windows, and investigate the impact of lockdown-induced structural breaks on model transferability across time.

REFERENCES

[1] GBD 2019 Risk Factors Collaborators, "Global burden of 87 risk factors in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019," *The Lancet*, vol. 396, no. 10258, pp. 1223–1249, 2020.

[2] X. Chen, H. Liu, and Y. Wang, "Lag structure of air pollution effects on daily hospital admissions: A meta-analysis of time-series studies," *Environmental Research*, vol. 228, p. 115831, 2023.

[3] S. Gulia, S. M. S. Nagendra, M. Khare, and I. Khanna, "Urban air quality management: A review," *Atmospheric Pollution Research*, vol. 6, no. 2, pp. 286–304, 2015.

[4] X. Hu, J. Belle, X. Meng, A. Wildani, L. Waller, M. Strickland, and Y. Liu, "Estimating PM2.5 concentrations in the conterminous United States using the random forest approach," *Environmental Science and Technology*, vol. 54, no. 21, pp. 13686–13694, 2020.

[5] M. T. Islam, A. I. Inan, and M. A. Hassan, "Non-parametric statistical approaches for air quality assessment in urban Bangladesh," *Journal of Environmental Science and Health, Part A*, vol. 57, no. 4, pp. 285–296, 2022.

[6] P. S. Rao and D. K. Khattar, "City-level heterogeneity in air quality risk: A chi-square analysis of Indian CPCB records," *Urban Climate*, vol. 38, p. 100901, 2021.

[7] R. Kumar and M. Goyal, "Temporal dynamics of air pollutants across Indian cities: An ADF-based stationarity assessment," *Atmospheric Environment*, vol. 201, pp. 102–113, 2019.

[8] J. S. Bedi and J. Dhankhar, "Machine learning-based air quality index prediction in Delhi using ensemble approaches," *Arabian Journal of Geosciences*, vol. 11, no. 22, p. 697, 2018.

[9] A. Verma, S. Kaur, and H. S. Sandhu, "KNN-based AQI classification for Indian cities: Impact of k selection and feature scaling," *Environmental Modeling and Assessment*, vol. 27, no. 3, pp. 511–523, 2022.

[10] D. Mishra and M. Goyal, "Predicting ambient PM2.5 in Delhi using regression-based models with multicollinearity-aware feature selection," *Atmospheric Research*, vol. 163, pp. 44–58, 2015.

[11] P. Sharma, V. Mahajan, and R. K. Gupta, "Deep learning for Indian air quality forecasting: When do LSTMs outperform classical ensembles?" *Computers and Geosciences*, vol. 186, p. 105576, 2024.

[12] A. K. Singh, N. Tiwari, and D. Kumar, "Statistical characterisation of PM2.5 inter-city variability in northern India using Kruskal-Wallis and Dunn post-hoc analysis," *Science of the Total Environment*, vol. 878, p. 163071, 2023.

[13] R. Jain, S. Mehrotra, and P. Arora, "Class imbalance in binary AQI risk classification: An empirical study on Indian monitoring data," *Environmental Data Science*, vol. 3, e14, 2024.